day_8_genome_assembly_Debarshi

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```
library(Rsubread)
setwd("/home/ibab/bio-env")
gtf_file <- "GCF_000146045.2_R64_genomic.gtf"

out_FeatCount_conA_rep1 <- featureCounts(
  files = "conA_rep1.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)
```

```
##
##          ==========     _____ _     _ ____  _____  _______          _____
##          =====         / ____| |  | |  _ \|  __ \|  ____|   /\    |  __ \
##            =====      | (___ | |  | | |_) | |__) | |__     /  \   | |  | |
##              ====      \___ \| |  | |  _ <|  _  /|  __|   / /\ \  | |  | |
##                 ====   ____) | |__| | |_) | | \ \| |____ / ____ \| |__| |
##          ==========   |_____/ \____/|____/|_|  \_\______/_/    \_\_____/
##         Rsubread 2.24.0
##
## //========================== featureCounts setting ===========================\\
## ||                                                                            ||
## ||             Input files : 1 SAM file                                       ||
## ||                                                                            ||
## ||                           conA_rep1.sam                                    ||
## ||                                                                            ||
## ||              Paired-end : no                                               ||
## ||        Count read pairs : no                                               ||
## ||              Annotation : GCF_000146045.2_R64_genomic.gtf (GTF)            ||
## ||      Dir for temp files : .                                                ||
## ||                 Threads : 1                                                ||
## ||                   Level : meta-feature level                               ||
## ||      Multimapping reads : counted                                          ||
## || Multi-overlapping reads : not counted                                      ||
## ||   Min overlapping bases : 1                                                ||
## ||                                                                            ||
## \\============================================================================//
##
## //================================ Running ===================================\\
## ||                                                                            ||
## || Load annotation file GCF_000146045.2_R64_genomic.gtf ...                   ||
## ||    Features : 6831                                                         ||
## ||    Meta-features : 6459                                                    ||
## ||    Chromosomes/contigs : 17                                                ||
```

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```
## ||                                                                            ||
## || Process SAM file conA_rep1.sam...                                          ||
## ||    Single-end reads are included.                                          ||
## ||    Total alignments : 800000                                               ||
## ||    Successfully assigned alignments : 721531 (90.2%)                       ||
## ||    Running time : 0.01 minutes                                             ||
## ||                                                                            ||
## || Write the final count table.                                               ||
## || Write the read assignment summary.                                         ||
## ||                                                                            ||
## \\============================================================================//
```

```
out_FeatCount_conA_rep2 <- featureCounts(
  files = "conA_rep2.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)
```

```
##
##          ==========     _____ _     _ ____  _____  _______          _____
##          =====         / ____| |  | |  _ \|  __ \|  ____|   /\    |  __ \
##            =====      | (___ | |  | | |_) | |__) | |__     /  \   | |  | |
##              ====      \___ \| |  | |  _ <|  _  /|  __|   / /\ \  | |  | |
##                ====    ____) | |__| | |_) | | \ \| |____ / ____ \| |__| |
##          ==========   |_____/ \____/|____/|_|  \_\______/_/    \_\_____/
##         Rsubread 2.24.0
##
## //========================== featureCounts setting ===========================\\
## ||                                                                            ||
## ||             Input files : 1 SAM file                                       ||
## ||                                                                            ||
## ||                           conA_rep2.sam                                    ||
## ||                                                                            ||
## ||              Paired-end : no                                               ||
## ||        Count read pairs : no                                               ||
## ||              Annotation : GCF_000146045.2_R64_genomic.gtf (GTF)            ||
## ||      Dir for temp files : .                                                ||
## ||                 Threads : 1                                                ||
## ||                   Level : meta-feature level                               ||
## ||      Multimapping reads : counted                                          ||
## || Multi-overlapping reads : not counted                                      ||
## ||   Min overlapping bases : 1                                               ||
## ||                                                                            ||
## \\============================================================================//
##
## //================================= Running ==================================\\
## ||                                                                            ||
## || Load annotation file GCF_000146045.2_R64_genomic.gtf ...                   ||
## ||    Features : 6831                                                         ||
## ||    Meta-features : 6459                                                    ||
## ||    Chromosomes/contigs : 17                                                ||
```

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```
## ||                                                                            ||
## || Process SAM file conA_rep2.sam...                                          ||
## ||    Single-end reads are included.                                          ||
## ||    Total alignments : 900000                                               ||
## ||    Successfully assigned alignments : 816269 (90.7%)                       ||
## ||    Running time : 0.01 minutes                                             ||
## ||                                                                            ||
## || Write the final count table.                                               ||
## || Write the read assignment summary.                                         ||
## ||                                                                            ||
## \\============================================================================//
```

```
out_FeatCount_conB_rep1 <- featureCounts(
  files = "conB_rep1.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)
```

```
##
##          ==========     _____ _     _ ____  _____  _______          _____
##          =====         / ____| |  | |  _ \|  __ \|  ____|   /\    |  __ \
##            =====      | (___ | |  | | |_) | |__) | |__     /  \   | |  | |
##              ====      \___ \| |  | |  _ <|  _  /|  __|   / /\ \  | |  | |
##                ====    ____) | |__| | |_) | | \ \| |____ / ____ \| |__| |
##          ==========   |_____/ \____/|____/|_|  \_\______/_/    \_\_____/
##         Rsubread 2.24.0
##
## //========================== featureCounts setting ===========================\\
## ||                                                                            ||
## ||             Input files : 1 SAM file                                       ||
## ||                                                                            ||
## ||                           conB_rep1.sam                                    ||
## ||                                                                            ||
## ||              Paired-end : no                                               ||
## ||        Count read pairs : no                                               ||
## ||              Annotation : GCF_000146045.2_R64_genomic.gtf (GTF)            ||
## ||      Dir for temp files : .                                                ||
## ||                 Threads : 1                                                ||
## ||                   Level : meta-feature level                               ||
## ||      Multimapping reads : counted                                          ||
## || Multi-overlapping reads : not counted                                      ||
## ||   Min overlapping bases : 1                                               ||
## ||                                                                            ||
## \\============================================================================//
##
## //================================ Running ===================================\\
## ||                                                                            ||
## || Load annotation file GCF_000146045.2_R64_genomic.gtf ...                   ||
## ||    Features : 6831                                                         ||
## ||    Meta-features : 6459                                                    ||
## ||    Chromosomes/contigs : 17                                                ||
```

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```
## ||                                                                            ||
## || Process SAM file conB_rep1.sam...                                          ||
## ||    Single-end reads are included.                                          ||
## ||    Total alignments : 700000                                               ||
## ||    Successfully assigned alignments : 633555 (90.5%)                       ||
## ||    Running time : 0.01 minutes                                             ||
## ||                                                                            ||
## || Write the final count table.                                               ||
## || Write the read assignment summary.                                         ||
## ||                                                                            ||
## \\============================================================================//
```

```
out_FeatCount_conB_rep2 <- featureCounts(
  files = "conB_rep2.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)
```

```
##
##          ==========     _____ _     _ ____  _____  _______          _____
##          =====         / ____| |  | |  _ \|  __ \|  ____|   /\    |  __ \
##            =====      | (___ | |  | | |_) | |__) | |__     /  \   | |  | |
##              ====      \___ \| |  | |  _ <|  _  /|  __|   / /\ \  | |  | |
##                ====    ____) | |__| | |_) | | \ \| |____ / ____ \| |__| |
##          ==========   |_____/ \____/|____/|_|  \_\______/_/    \_\_____/
##         Rsubread 2.24.0
##
## //========================== featureCounts setting ===========================\\
## ||                                                                            ||
## ||             Input files : 1 SAM file                                       ||
## ||                                                                            ||
## ||                           conB_rep2.sam                                    ||
## ||                                                                            ||
## ||              Paired-end : no                                               ||
## ||        Count read pairs : no                                               ||
## ||              Annotation : GCF_000146045.2_R64_genomic.gtf (GTF)            ||
## ||      Dir for temp files : .                                                ||
## ||                 Threads : 1                                                ||
## ||                   Level : meta-feature level                               ||
## ||      Multimapping reads : counted                                          ||
## || Multi-overlapping reads : not counted                                      ||
## ||   Min overlapping bases : 1                                               ||
## ||                                                                            ||
## \\============================================================================//
##
## //================================ Running ===================================\\
## ||                                                                            ||
## || Load annotation file GCF_000146045.2_R64_genomic.gtf ...                   ||
## ||    Features : 6831                                                         ||
## ||    Meta-features : 6459                                                    ||
## ||    Chromosomes/contigs : 17                                                ||
```

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```
## ||                                                                      ||
## || Process SAM file conB_rep2.sam...                                    ||
## ||    Single-end reads are included.                                    ||
## ||    Total alignments : 1000000                                        ||
## ||    Successfully assigned alignments : 903130 (90.3%)                 ||
## ||    Running time : 0.01 minutes                                       ||
## ||                                                                      ||
## || Write the final count table.                                         ||
## || Write the read assignment summary.                                   ||
## ||                                                                      ||
## \\======================================================================//
```

```
out_FeatCount_conA_rep1_count = data.frame(out_FeatCount_conA_rep1$counts)
out_FeatCount_conA_rep2_count = data.frame(out_FeatCount_conA_rep2$counts)
out_FeatCount_conB_rep1_count = data.frame(out_FeatCount_conB_rep1$counts)
out_FeatCount_conB_rep2_count = data.frame(out_FeatCount_conB_rep2$counts)
```

```
count_matrix <- cbind(
out_FeatCount_conA_rep1_count,
out_FeatCount_conA_rep2_count,
out_FeatCount_conB_rep1_count,
out_FeatCount_conB_rep2_count
)
write.table(count_matrix, file = "count_matrix.txt",
sep="\t")
```

```
library(edgeR)
```

```
## Warning: package 'edgeR' was built under R version 4.5.3
```

```
## Loading required package: limma
```

```
## Loading required package: limma
group=c(rep("conA",2),rep("conB",2))
```

```
y=DGEList(counts=count_matrix,group=group)
y
```

```
## An object of class "DGEList"
## $counts
##           conA_rep1.sam conA_rep2.sam conB_rep1.sam conB_rep2.sam
## YAL068C               0             0             0             0
## YAL067W-A             0             0             0             0
## YAL067C               0             0             0             0
## YAL065C               0             0             0             0
## YAL064W-B             1             0             1             0
## 6454 more rows ...
##
```

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```
## $samples
##                group lib.size norm.factors
## conA_rep1.sam  conA   721531            1
## conA_rep2.sam  conA   816269            1
## conB_rep1.sam  conB   633555            1
## conB_rep2.sam  conB   903130            1
```

```
# Filter lowly expressed genes #filterByExpr() identifies genes with sufficient express
ion to be reliably tested. Applies the cpm and gives cutoff
keep=filterByExpr(y)
head(keep)
```

```
##   YAL068C YAL067W-A   YAL067C   YAL065C YAL064W-B YAL064C-A
##     FALSE     FALSE     FALSE     FALSE     FALSE     FALSE
```

```
# how many genes survived
sum(keep)
```

```
## [1] 5224
```

```
# to see the total number of genes before filtering.
length(keep)
```

```
## [1] 6459
```

```
library(edgeR)

# 1. Ensure design matrix is defined first
design1 <- model.matrix(~0 + group)

# 2. Recalculate 'keep' using filterByExpr on object 'y'
keep <- filterByExpr(y, design = design1)

# 3. Filter the DGEList (lengths will now match)
y <- y[keep, , keep.lib.sizes = FALSE]
dim(y)
```

```
## [1] 5224    4
```

```
# 4. Calculate normalization factors (TMM)
y <- calcNormFactors(y)

# 5. Estimate dispersion
y <- estimateDisp(y, design1)

# 6. Test for differential expression
# Note: exactTest requires treatment group factors; ensure group is set on y$samples$gr
oup
```

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```
DEet <- exactTest(y, pair = c("conA", "conB"))

# View top results
head(DEet$table)
```

```
##              logFC   logCPM     PValue
## YAL063C -0.7996891 3.871542 0.17528734
## YAL062W -0.7794297 4.791989 0.10740418
## YAL061W -0.8961737 8.968509 0.04103521
## YAL060W -0.7463570 9.689610 0.04324005
## YAL059W  1.0540346 4.477723 0.05977212
## YAL058W  0.4713354 5.690378 0.13805991
```

```
#Identifying the up and down regulated gene at least 2-fold higher expression and stron
g statistical significance
et_table = DEet$table
up_et = subset(et_table,et_table$logFC >= 1 & et_table$PValue <= 0.001)
down_et = subset(et_table,et_table$logFC <= -1 & et_table$PValue <= 0.001)
```

```
# up regulated gene
head(up_et)
```

```
##            logFC    logCPM       PValue
## YAL025C 2.725719  5.599999 3.113077e-13
## YBL042C 1.930598  6.457424 2.155048e-11
## YBL039C 1.959675  6.764031 8.795183e-08
## YBL024W 1.488132  6.681218 1.595984e-04
## YBL004W 1.130756  6.618267 4.097396e-04
## YBR031W 1.453359 11.303596 9.933219e-04
```

```
# Downregulated genes
head(down_et)
```

```
##               logFC   logCPM       PValue
## YAL008W   -1.311142 6.210347 4.367562e-06
## YBL064C   -1.235612 8.116882 2.765428e-04
## YBL015W   -1.400856 8.964937 1.640541e-04
## YBR053C   -1.087734 7.310590 9.051842e-04
## YBR072W   -3.510040 8.674449 2.186622e-12
## YBR089C-A -1.232368 5.891167 4.559913e-05
```

```
#No of gene upregulated and down regulated
nrow(up_et)
```

```
## [1] 235
```

```
nrow(down_et)
```

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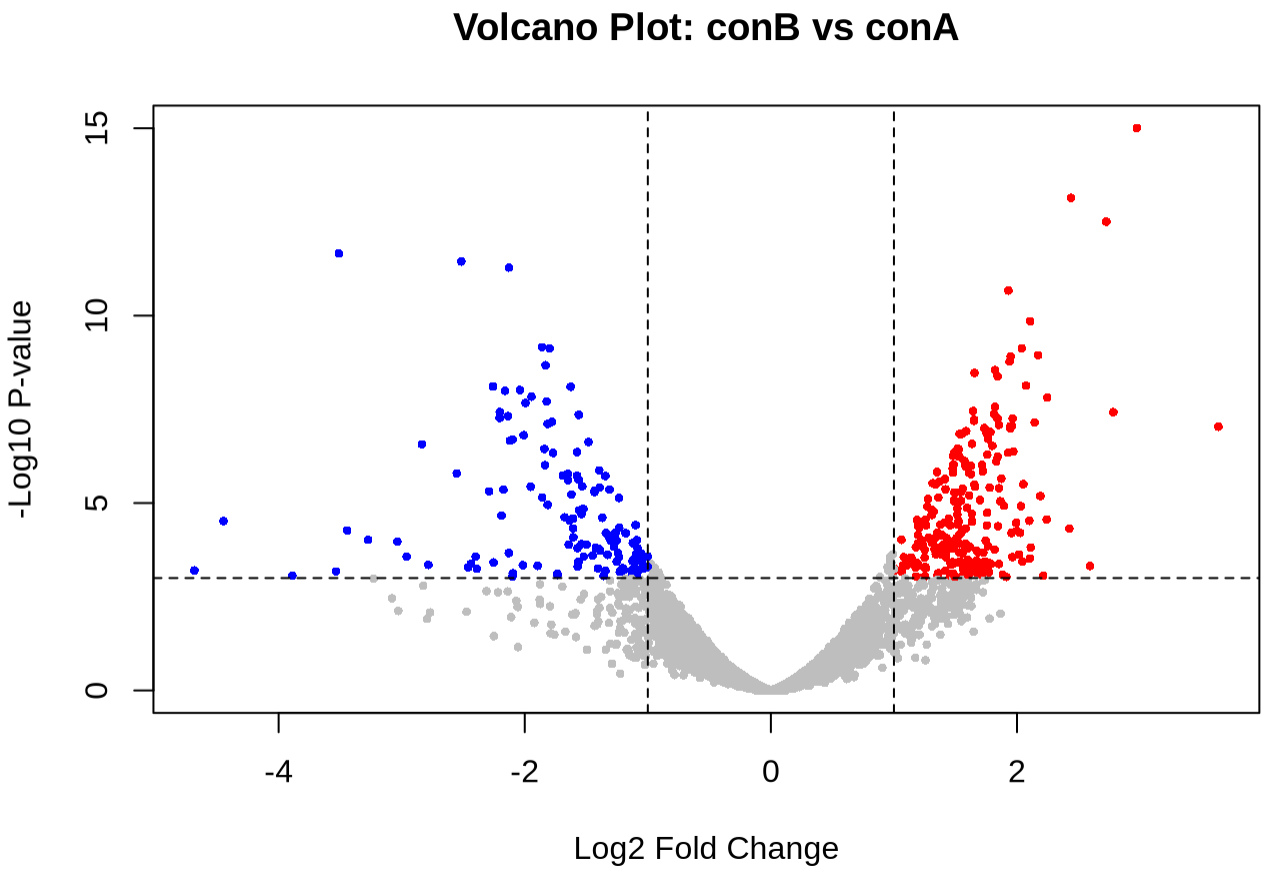
```
## [1] 135
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.3
```

```
# Create a color column
et_table$color <- "grey"
# Upregulated genes
et_table$color[et_table$logFC >= 1 & et_table$PValue <= 0.001] <- "red"
# Downregulated genes
et_table$color[et_table$logFC <= -1 & et_table$PValue <= 0.001] <- "blue"
# Calculate -log10(P-value)
et_table$negLogP <- -log10(et_table$PValue)
# Volcano plot
plot(
et_table$logFC,
et_table$negLogP,
pch = 16,
cex = 0.6,
col = et_table$color,
xlab = "Log2 Fold Change",
ylab = "-Log10 P-value",
main = "Volcano Plot: conB vs conA"
)
# Vertical cutoff lines
abline(
v = c(-1, 1),
lty = 2,
col = "black"
)
# Horizontal cutoff line
abline(
h = -log10(0.001),
lty = 2,
col = "black"
)
```

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```
# RPKM_Normalization for raw read counts
gene_length=out_FeatCount_conA_rep1$annotation$Length
head(gene_length)
```

```
## [1]  363  228 1782  387  381  381
```

```
gene_id <- out_FeatCount_conA_rep1$annotation$GeneID
head(gene_id)
```

```
## [1] "YAL068C"   "YAL067W-A" "YAL067C"   "YAL065C"   "YAL064W-B" "YAL064C-A"
```

```
gene_info <- data.frame(Geneid = gene_id,Length = gene_length)
head(gene_info)
```

```
##      Geneid Length
## 1   YAL068C    363
## 2 YAL067W-A    228
## 3   YAL067C   1782
## 4   YAL065C    387
## 5 YAL064W-B    381
## 6 YAL064C-A    381
```

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```
gene_info <- gene_info[match(rownames(count_matrix), gene_info$Geneid),]
count_data <- cbind(
Geneid = rownames(count_matrix),
Length = gene_info$Length,
count_matrix,
row.names = NULL
)
write.table(count_data, file = "count_data.txt",
sep="\t")
dim(count_data)
```

```
## [1] 6459    6
```

```
library_size <- colSums(count_data[, 3:6])
library_size
```

```
## conA_rep1.sam conA_rep2.sam conB_rep1.sam conB_rep2.sam
##        721531        816269        633555        903130
```

```
scaling_factor <- library_size / 1000000
scaling_factor
```

```
## conA_rep1.sam conA_rep2.sam conB_rep1.sam conB_rep2.sam
##      0.721531      0.816269      0.633555      0.903130
```

```
gene_length_kb <- count_data$Length / 1000
#Calculate RPKM
RPKM_conA_rep1<-count_data$conA_rep1.sam/(gene_length_kb*scaling_factor[1])
RPKM_conA_rep2<-count_data$conA_rep2.sam/(gene_length_kb*scaling_factor[2])
RPKM_conB_rep1<-count_data$conB_rep1.sam/(gene_length_kb*scaling_factor[3])
RPKM_conB_rep2<-count_data$conB_rep2.sam/(gene_length_kb*scaling_factor[4])
#Create the RPKM matrix
rpkm <- data.frame(
Geneid = count_data$Geneid,
RPKM_conA_rep1 = RPKM_conA_rep1,
RPKM_conA_rep2 = RPKM_conA_rep2,
RPKM_conB_rep1 = RPKM_conB_rep1,
RPKM_conB_rep2 = RPKM_conB_rep2
)
rownames(rpkm) <- rpkm$Geneid
rpkm$Geneid <- NULL
#Log transform for heatmap
log_rpkm <- log2(rpkm + 1)
```

```
#For DEG list perform RPKM Normalization
up_genes <- rownames(up_et)
down_genes <- rownames(down_et)
```

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```
deg_genes <- c(up_genes, down_genes)
length(deg_genes)
```

```
## [1] 370
```

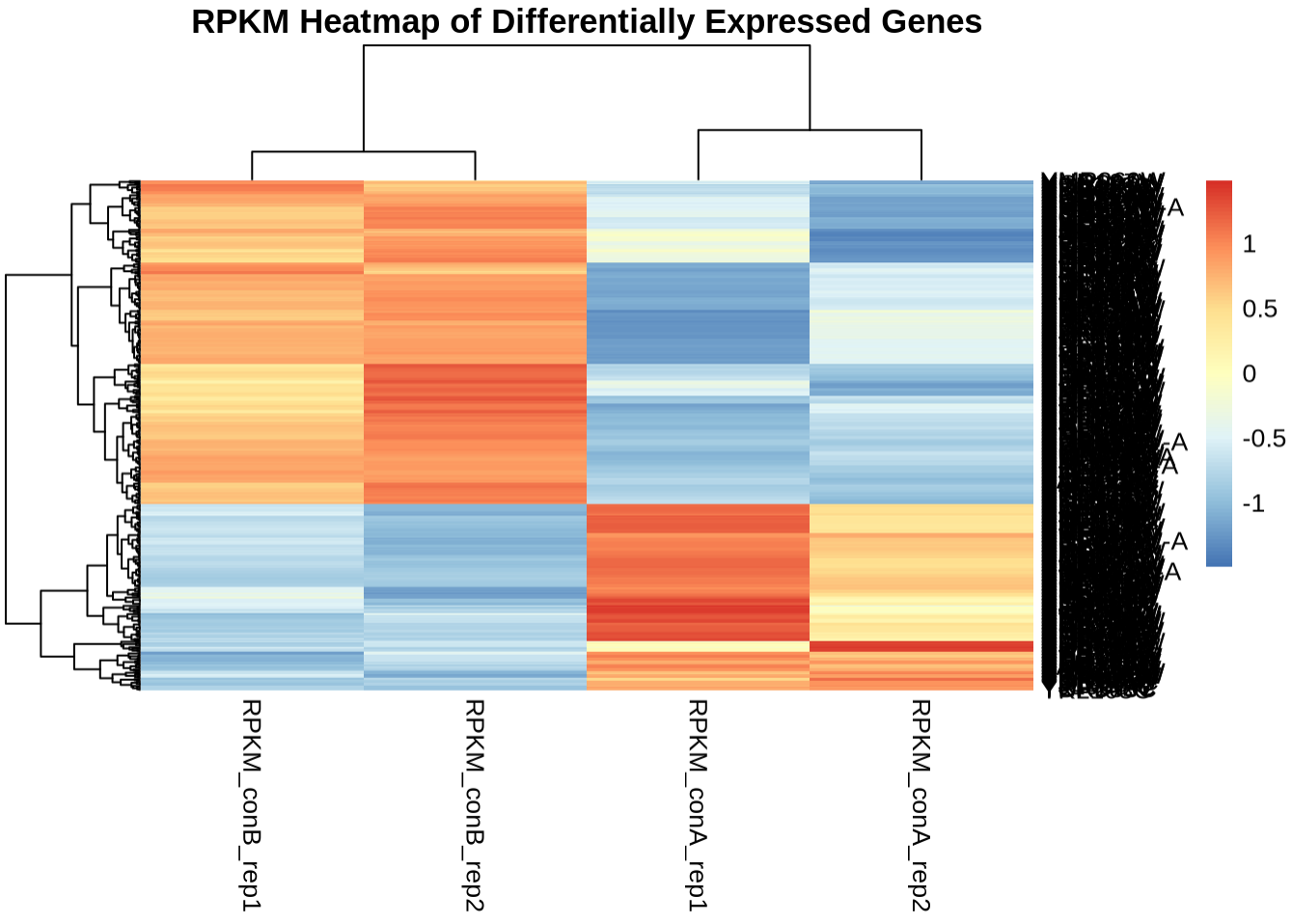
```
rpkm_deg <- rpkm[deg_genes, ]
dim(rpkm_deg)
```

```
## [1] 370   4
```

```
log_rpkm_deg <- log2(rpkm_deg + 1)
library(pheatmap)
```

```
## Warning: package 'pheatmap' was built under R version 4.5.3
```

```
pheatmap(
log_rpkm_deg,
scale = "row",
cluster_rows = TRUE,
cluster_cols = TRUE,
main = "RPKM Heatmap of Differentially Expressed Genes"
)
```



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```
# Quasi-likelihood method

group=c(rep("conA",2),rep("conB",2))
group
```

```
## [1] "conA" "conA" "conB" "conB"
```

```
#DGEList is an edgeR object that stores your RNA-seq count data along with information
about the experim
y=DGEList(counts=count_matrix,group=group)
keep=filterByExpr(y)
y=y[keep,,keep.lib.sizes=FALSE]
dim(y)
```

```
## [1] 5224    4
```

```
y <- calcNormFactors(y)
design=model.matrix(~0+group)
design
```

```
##   groupconA groupconB
## 1         1         0
## 2         1         0
## 3         0         1
## 4         0         1
## attr(,"assign")
## [1] 1 1
## attr(,"contrasts")
## attr(,"contrasts")$group
## [1] "contr.treatment"
```

```
# This fits a quasi-likelihood negative binomial generalized linear model to your RNA-s
eq count data.
y_glm=glmQLFit(y,design)
BvsA=makeContrasts(groupconB - groupconA, levels=design)
#This performs the actual quasi-likelihood F-test for the contrast:
res = glmQLFTest(y_glm, contrast = BvsA)
#topTags(res)

Ftest_table <- res$table
up_Ftest <- subset(Ftest_table,logFC >= 1 & PValue <= 0.01)
down_Ftest <- subset(Ftest_table,logFC <= -1 & PValue <= 0.01)
nrow(up_Ftest)
```

```
## [1] 255
```

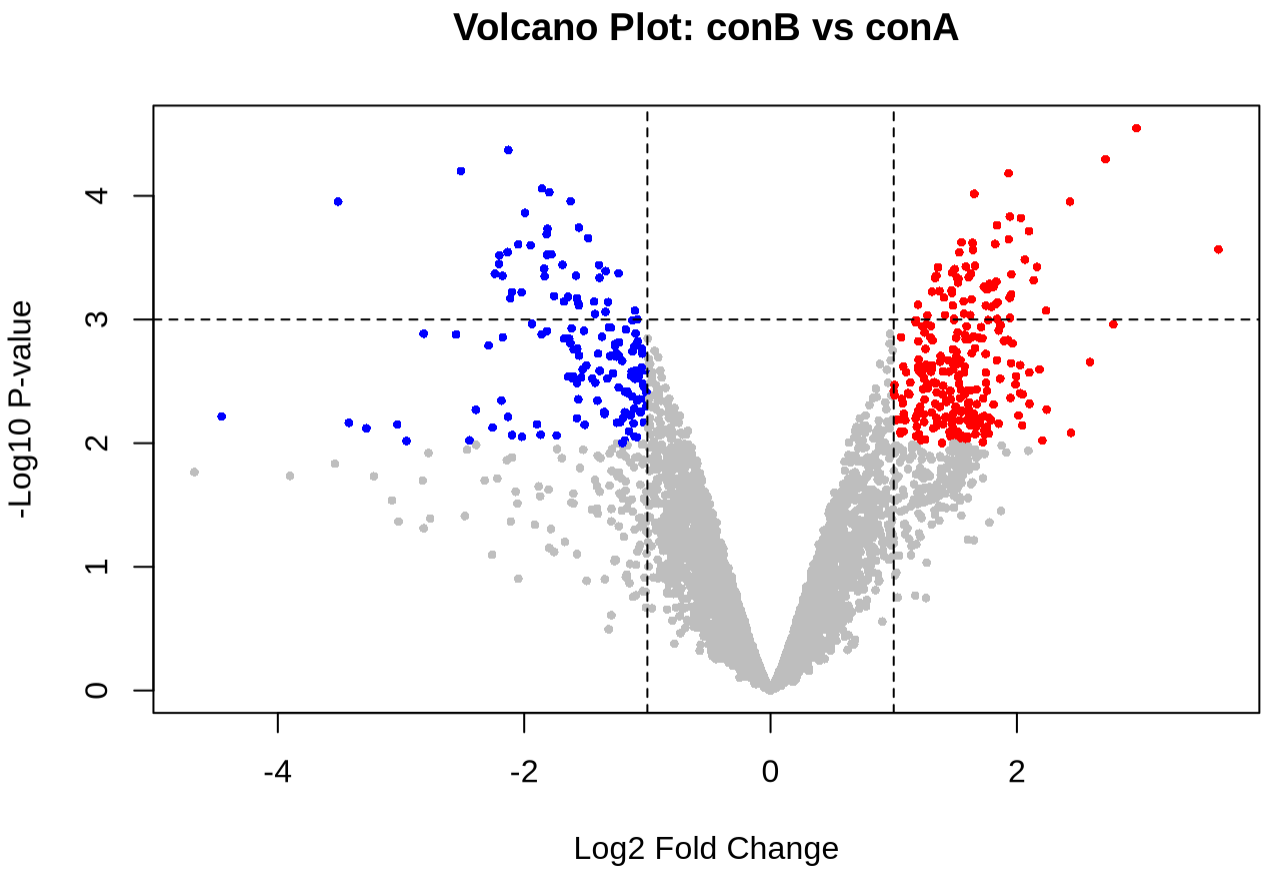
```
nrow(down_Ftest)
```

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```
## [1] 148
```

```
Ftest_table$color <- "grey"
# Upregulated genes
Ftest_table$color[Ftest_table$logFC >= 1 & Ftest_table$PValue <= 0.01] <- "red"
# Downregulated genes
Ftest_table$color[Ftest_table$logFC <= -1 & Ftest_table$PValue <= 0.01] <- "blue"
# Calculate -log10(P-value)
Ftest_table$negLogP <- -log10(Ftest_table$PValue)
# Volcano plot
plot(
Ftest_table$logFC,
Ftest_table$negLogP,
pch = 16,
cex = 0.6,
col = Ftest_table$color,
xlab = "Log2 Fold Change",
ylab = "-Log10 P-value",
main = "Volcano Plot: conB vs conA"
)
# Vertical cutoff lines
abline(
v = c(-1, 1),
lty = 2,
col = "black"
)
# Horizontal cutoff line
abline(
h = -log10(0.001),
lty = 2,
col = "black"
)
```

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